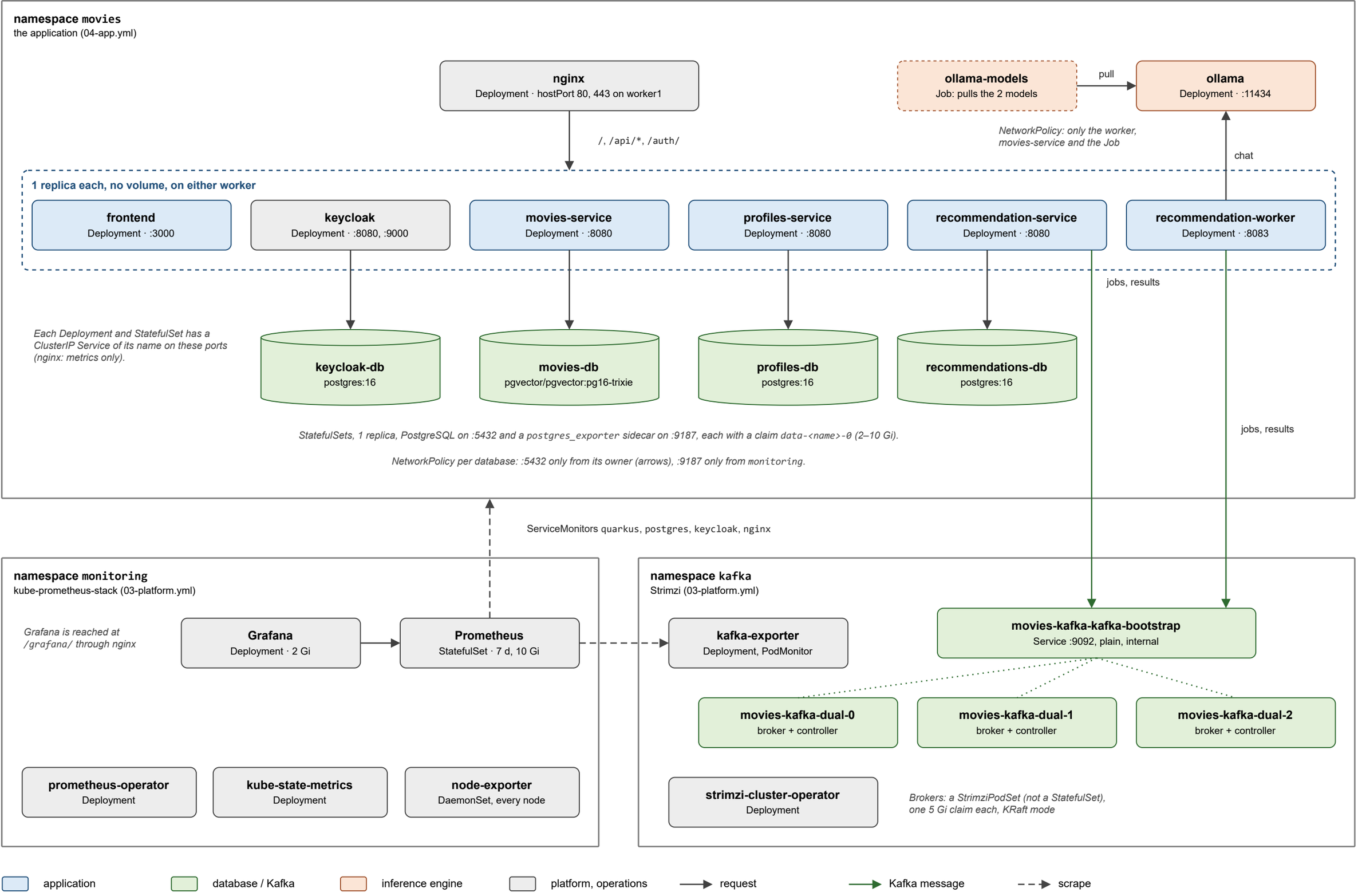

namespace movies
the application (04-app.yml)
nginx
Deployment · hostPort 80, 443 on worker1
/, /api/*, /auth/
ollama-models
Job: pulls the 2 models
pull
ollama
Deployment · :11434
NetworkPolicy: only the worker, movies-service and the Job
chat
1 replica each, no volume, on either worker
frontend
Deployment · :3000
keycloak
Deployment · :8080, :9000
movies-service
Deployment · :8080
profiles-service
Deployment · :8080
recommendation-service
Deployment · :8080
recommendation-worker
Deployment · :8083
jobs, results
Each Deployment and StatefulSet has a ClusterIP Service of its name on these ports (nginx: metrics only).
keycloak-db
postgres:16
movies-db
pgvector/pgvector:pg16-trixie
profiles-db
postgres:16
recommendations-db
postgres:16
jobs, results
StatefulSets, 1 replica, PostgreSQL on :5432 and a postgres_exporter sidecar on :9187, each with a claim data-<name>-0 (2–10 Gi).
NetworkPolicy per database: :5432 only from its owner (arrows), :9187 only from monitoring.
ServiceMonitors quarkus, postgres, keycloak, nginx
namespace monitoring
kube-prometheus-stack (03-platform.yml)
Grafana is reached at /grafana/ through nginx
Grafana
Deployment · 2 Gi
Prometheus
StatefulSet · 7 d, 10 Gi
prometheus-operator
Deployment
kube-state-metrics
Deployment
node-exporter
DaemonSet, every node
namespace kafka
Strimzi (03-platform.yml)
kafka-exporter
Deployment, PodMonitor
movies-kafka-kafka-bootstrap
Service :9092, plain, internal
movies-kafka-dual-0
broker + controller
movies-kafka-dual-1
broker + controller
movies-kafka-dual-2
broker + controller
strimzi-cluster-operator
Deployment
Brokers: a StrimziPodSet (not a StatefulSet), one 5 Gi claim each, KRaft mode
application
database / Kafka
inference engine
platform, operations
request
Kafka message
scrape